



NILUS LAB

Reversing Cardiac Ageing Without Oncogene Risk

Generative cellular reprogramming therapies targeting cardiomyopathy and heart failure.
Powered by the Zenith v28.0 GOLD single-cell foundation model.

DEEP BIO ACCELERATOR - LAUNCH TRACK INTERVIEW PRESENTATION

CLINICAL RISK & LIMITS

Cardiac Senescence & Reprogramming Limits

- **64M+ Patients Globally:**
Heart failure is driven by cellular senescence, loss of adult sarcomere structures, and chromatin degradation.
- **Oncogenic Risk (OSKM):**
Classical Yamanaka factors trigger dedifferentiation and cell drift, creating high risks of teratoma and tumor formation.
- **Delivery & Specificity Limits:**
Existing therapies cannot precisely target therapeutic chromatin states, leading to unsafe, non-specific changes.

ZENITH PLATFORM SOLUTION

Safe, Target-Specific Rejuvenation

- **In-Silico Safety Gating:**
Dual-threshold gates screen out oncogenes and restrict cellular dedifferentiation past functional limits.
- **-11.9y Epigenetic Reset:**
Validated age clock trained on real HCA data achieves 11.9 years of age reversal in mature cells.
- **Transient Cooperativity:**
Optimized expression windows (e.g., 2h ON) regulate chromatin accessibility without altering cell identity.

1 scVI Generative Core

- Variational Autoencoder (scVI) maps cellular manifolds inside a 20-dimensional latent space.
- Trained on 500k real human cardiac cells across 4,908 Highly Variable Gene (HVG) dimensions.
- Models chromatin accessibility states to predict cell rejuvenation without lineage loss.

2 Safety Auditing

- Oncogene Blacklist: Screens transcription factor candidates to eliminate MYC/OCT4-related tumor risks.
- Dedifferentiation Gating: Limits changes to keep structural markers (like TNNT2/ACTN2) stable.
- Restricts cell states from entering uncontrolled pluripotency.

3 AF3 & ElasticNet Clock

- AlphaFold 3 Valency: Predicts cooperative transcription factor-DNA binding coordinate mechanics.
- ElasticNet Clock: Regresses latent scVI spaces against donor age. Yields -11.9y age reset.
- Robust Trust: Achieves a cross-validated Mean Absolute Error (MAE) of 6.0 years on real HCA donors.

MARKET POSITIONING & COMPETITION**Target Market, Competition & Differentiation**

- Indication Strategy: Focused on Cardiomyopathy and Heart Failure indication targets (\$50B longevity & CVD market indication).
- Competition (Altos/Retro): Giant-scale ventures target broad, systemic multi-tissue rejuvenations (high cost). Nilus targets high-specificity cardiac restoration.
- Differentiation (Turn Bio): Focuses on skin/dermatology. Nilus specializes in the cardiac lineage utilizing VAE safety ceilings to block cell state drift.
- Business Model: Out-license in-silico vetted cocktails to pharma.

WETLAB VALIDATION FUNDING ASK

Raising a \$1.5M Seed Round to fund wetlab translation: TF cocktail synthesis, calcium-transient imaging assays, chromatin accessibility profiling, and rodent safety trials.

\$1.5M Seed

WETLAB FUNDING REQUIREMENT

-11.9 Years

EPIGENETIC AGE RESET (MAE=6.0y)

THE TEAM

Team Chemistry & Dynamics

- Interdisciplinary Founders: Unifies expertise across VAE computational deep learning, software engineering, and wetlab experimental molecular biology.
- Long-standing Research Partnership: The co-founders have a history of collaborative research in single-cell transcriptomics.
- Interview Preparedness: Scheduled video call to align on Zenith's technical architecture, safety thresholds, and program goals.

DEVELOPMENT ROADMAP

Key Execution Milestones



Phase 1: Cocktail Synthesis & Wetlab Setup

Synthesize predicted transcription factor cocktails and set up human primary cell lines.



Phase 2: Preclinical In-Vitro Assaying

Measure chromatin accessibility changes and track contractility and calcium transient dynamics.



Phase 3: In-Vivo Rodent Assays & Seed Closing

Conclude a \$1.5M Seed round to finance mouse-model cardiomyopathy and heart failure safety trials.